

Scenario 1: Identify Consistent High Performers

Question: Find employees who have consistently received a salary increase for the past 3 years.

Answer:

- ☐ Select the **employee** table and **salary_history** table.
- ☐ Join the salary history table three times to retrieve salary records for the last three years.
- ☐ Use the **WHERE** clause to compare salaries year by year.
- ☐ The employee is considered a consistent high performer if:
 - Salary in 2024 > Salary in 2023
 - Salary in 2025 > Salary in 2024

```
SELECT e.Name
FROM employee e
JOIN salary_history s1 ON e.EmployeeID = s1.EmployeeID AND s1.Year = 2023
JOIN salary_history s2 ON e.EmployeeID = s2.EmployeeID AND s2.Year = 2024
JOIN salary_history s3 ON e.EmployeeID = s3.EmployeeID AND s3.Year = 2025
WHERE s2.Salary > s1.Salary AND s3.Salary > s2.Salary;
```

Scenario 2: Customer Retention Analysis

Question: Find customers who made purchases in the last 6 months but not in the previous 6 months.

Answer:

- Select the **customer** and **orders** tables.
- Join both tables using CustomerID.
- Use the WHERE clause to find customers who placed orders in the **last 6 months**.
- Use the NOT IN subquery to exclude customers who made purchases in the **previous 6 months** (from 12 months ago up to 6 months ago).
- The result shows customers who are **newly active** in the last 6 months.

```
SELECT DISTINCT c.CustomerName
FROM customer c
```

```
JOIN order o ON c.CustomerID = o.CustomerID

WHERE o.OrderDate >= Date_SUB(CURDATE(), INTERVAL 6 MONTH)

AND c.CustomerID NOT IN (

    SELECT CustomerID

    FROM orders

    WHERE OrderDate >=Date_SUB(CURDATE(), INTERVAL 12 MONTH)

    AND OrderDate < DATE_SUB(CURDATE(), INTERVAL 6 MONTH));
```

Scenario 3: Identify Products with Declining Sales

Question: Find products whose sales decreased by more than 30% compared to the previous month.

Answer:

- Select the Products and Sales tables.
- Join the Sales table twice to compare the current month's sales with the previous month's sales for the same product.
- Filter the required months using the WHERE clause.
- Compare the current month's sales with 70% of the previous month's sales.
- If the current month's sales are less than 70% of the previous month's sales, it means sales decreased by more than 30%, so return that product.

```
SELECT P.Name

FROM Products P

JOIN Sales s1 ON P.ProductID = s1. ProductID

JOIN Sales s2 ON p.ProductID = s2.ProductID

WHERE s1.Month = '2026-05' AND s2.Month = '2026-06'

AND s2.TotalSales < s1.TotalSales * 0.70;
```

Scenario 4: Calculate Moving Average of Sales

Question: Calculate a 3-month moving average for product sales.

Answer:

- Select the product ID, month, and sales.
- Use the AVG() window function to calculate the average sales.
- PARTITION BY ProductID calculates the moving average separately for each product.
- ORDER BY Month ensures the rows are processed in chronological order.
- ROWS BETWEEN 2 PRECEDING AND CURRENT ROW creates a 3-month window consisting of the current month and the previous two months.
- The result is the 3-month moving average for each product.

```
SELECT ProductID, Month, Sales
```

```
    AVG(Sales) OVER (
```

```
        PARTITION BY ProductID
```

```
        ORDER BY Month
```

```
        ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
```

```
    ) AS MovingAverage
```

```
FROM Sales;
```

Scenario 5: Detect Duplicate Records

Question: Find all duplicate records from a customer table based on email and phone number.

Answer:

- Select the **Email** and **Phone** columns from the Customer table.
- Use **GROUP BY Email, Phone** to group customers having the same email and phone number.
- Use **COUNT(*)** to count how many records exist in each group.
- Use **HAVING COUNT(*) > 1** to return only those groups where the count is greater than 1.
- The result displays duplicate email and phone combinations.

```
SELECT Email, Phone,  
        COUNT(*) AS Duplicate_Count  
FROM Customer  
GROUP BY Email, Phone  
HAVING COUNT (*) > 1;
```

Scenario 6: Identify Products with No Sales for 3 Consecutive Months

Question: List products that have not had any sales for 3 consecutive months.

Answer:

- Select product names from the Products table.
- The subquery finds all products that had sales in **February, March, or April**.
- NOT IN excludes those products.
- The remaining products are those with **no sales during all three months**.

```
SELECT p.ProductName  
FROM Products p  
WHERE p.ProductID NOT IN (  
    SELECT ProductID  
    FROM Sales  
    WHERE Month IN ('2026-02', '2026-03', '2026-04')  
);
```
